

Audience Sensitivity, Content Compliance & Format Preference Analysis of France Top 50 Playlist

Research Paper

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Abstract

This research analyzes the France Top 50 playlist using Python, Pandas, Plotly, Matplotlib, Seaborn, and Streamlit to evaluate audience sensitivity, content compliance, and release format preferences. Interactive analytics and KPI dashboards provide business insights for music labels and streaming platforms.

This section explains the analytical approach in greater detail. The project emphasizes reproducible data preparation, exploratory data analysis, interactive visualization, KPI monitoring, and business interpretation. These methods can be reused for playlist analysis in other countries or genres.

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Introduction

Music streaming generates valuable behavioral data. This project studies popularity, explicit content, album type, artist presence, and song duration within France's Top 50 playlist.

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Problem Statement

Organizations need a data-driven method to understand listener preferences, explicit content performance, and release strategies.

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Objectives

Analyze playlist characteristics, calculate KPIs, visualize trends, and develop an interactive dashboard.

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Dataset Description

The dataset contains date, position, song, artist, popularity, duration_ms, album_type, total_tracks, is_explicit, and album_cover_url. Cleaning included datatype conversion, duplicate removal, and creation of duration_min.

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Methodology

Workflow: data collection, preprocessing, feature engineering, EDA, KPI calculation, dashboard development, interpretation.

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Exploratory Data Analysis

Charts included donut charts, histograms, scatter plots, bar charts, box plots, line charts, and a correlation heatmap.

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KPIs

KPIs include Total Songs, Unique Artists, Average Popularity, Average Duration, Explicit %, Album %, Single %, and Average Album Size.

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Dashboard Design

A Streamlit dashboard with filters, KPI cards, interactive Plotly charts, download option, and business insights was developed.

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Results and Discussion

The analysis reveals trends in popularity, duration, artist concentration, and explicit content distribution. Interactive filtering allows flexible exploration.

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Business Insights

The dashboard highlights market preferences, top artists, content mix, and structural characteristics of successful songs.

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Recommendations

Expand the solution using Spotify APIs, predictive models, and recommendation systems.

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Future Scope

Include machine learning, sentiment analysis, cloud deployment, and real-time streaming analytics.

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Conclusion

The project demonstrates how descriptive analytics and dashboards can support music industry decisions.

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References

Streamlit, Plotly, Pandas, Python, Spotify Developer documentation.